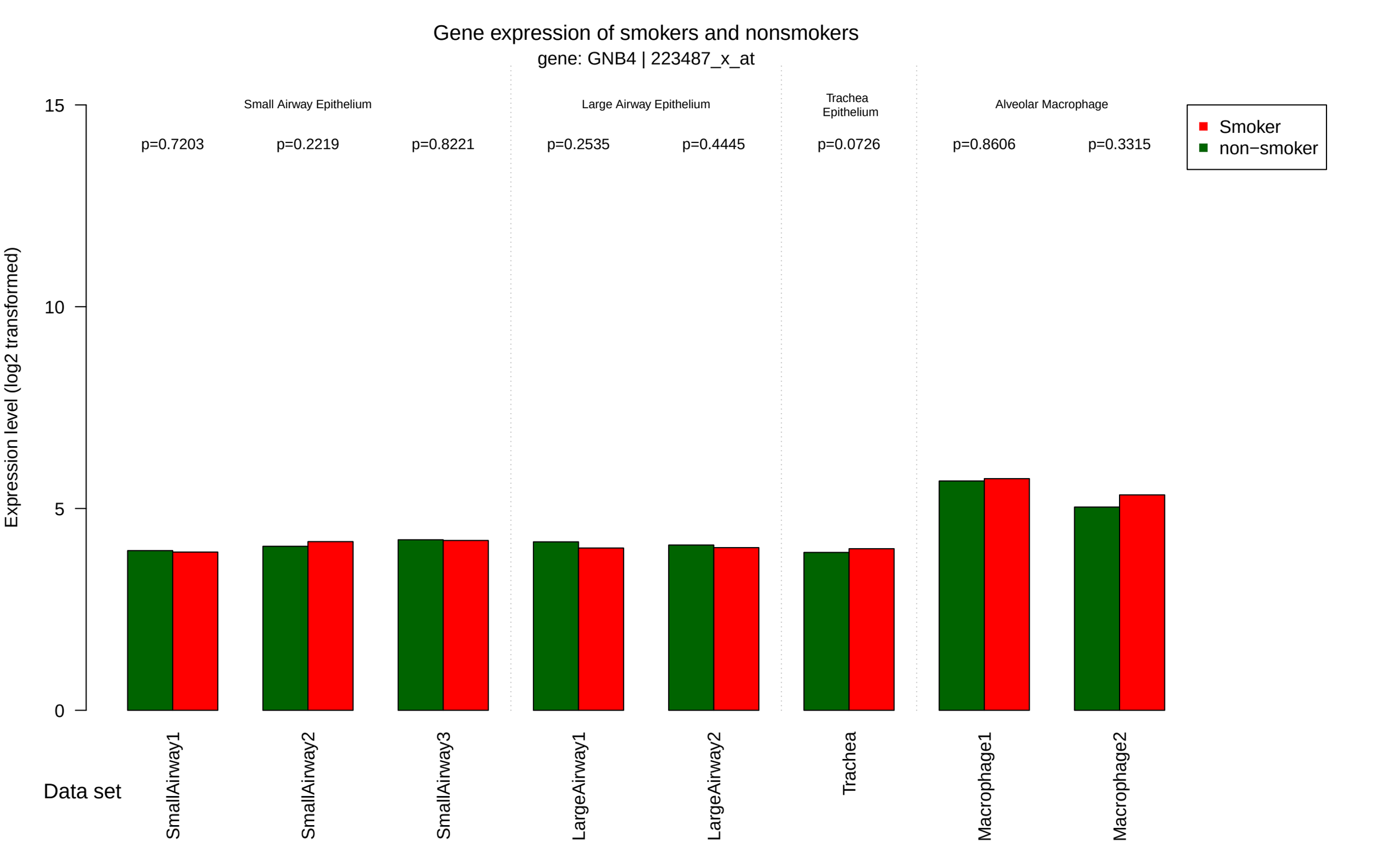
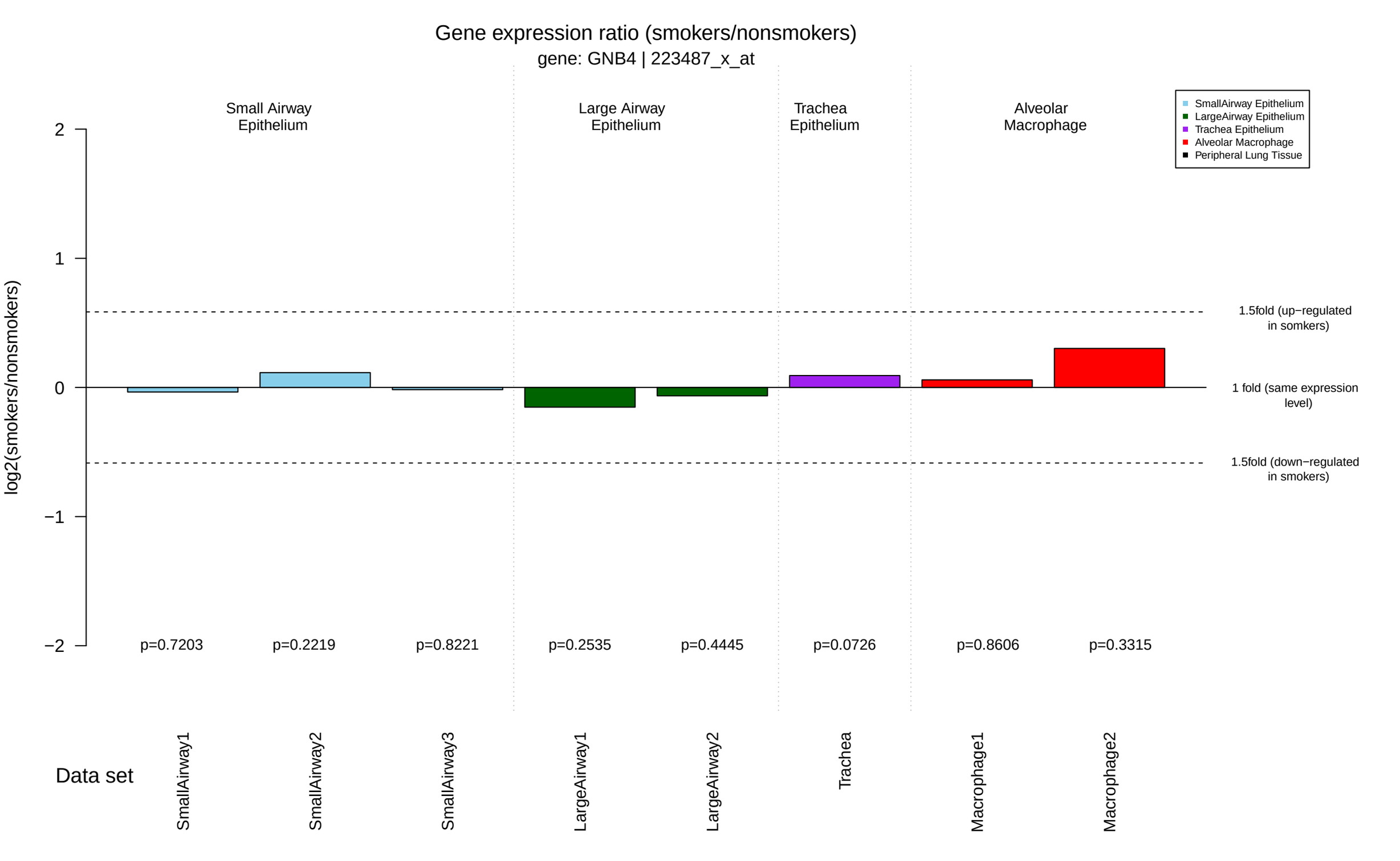
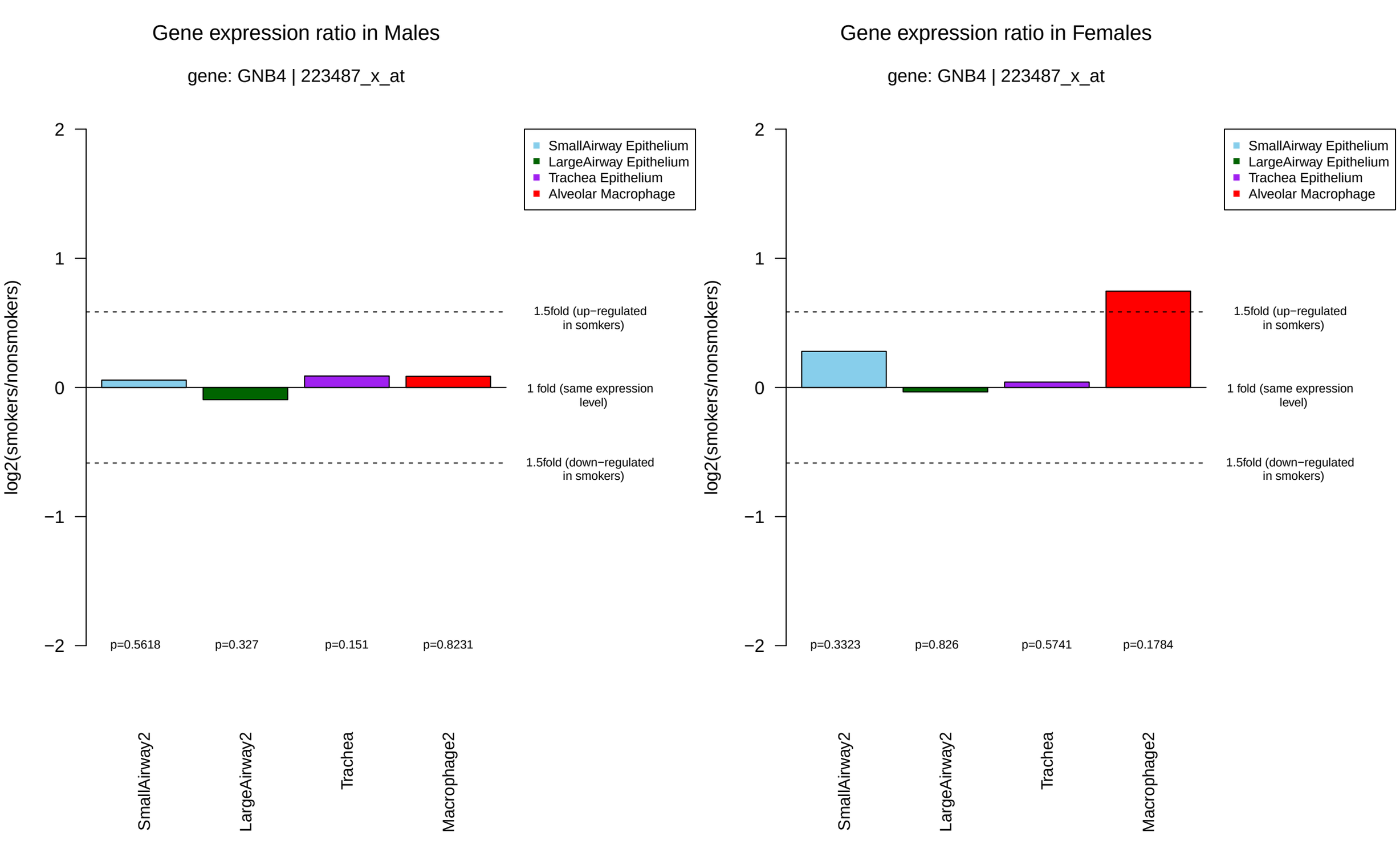
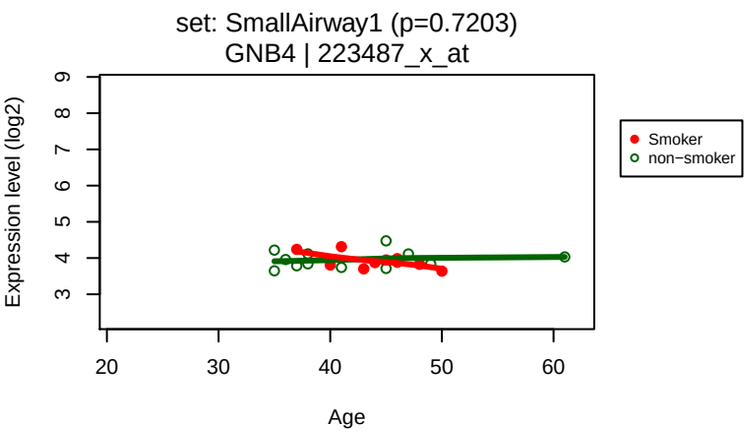
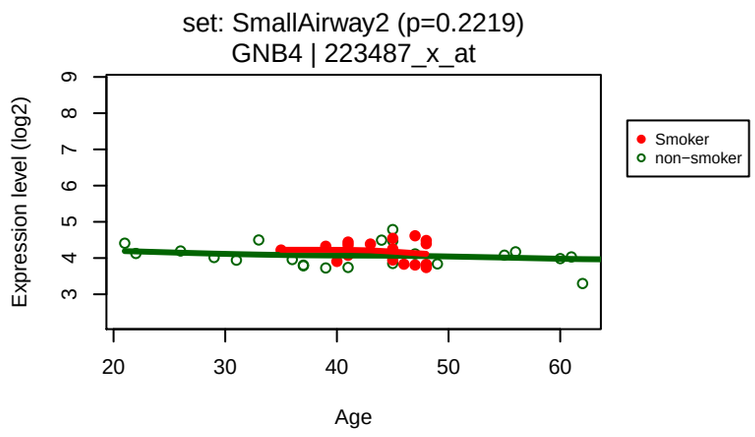
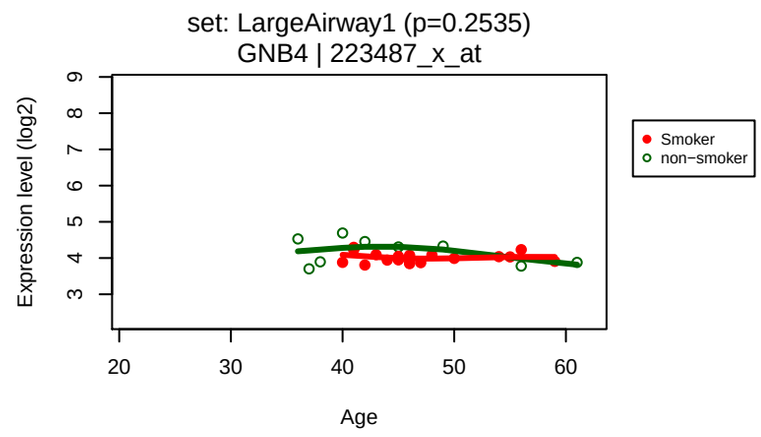

Gene expression of smokers and nonsmokers
gene: GNB4 | 223487_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7203
p=0.2219
p=0.8221
p=0.2535
p=0.4445
p=0.0726
p=0.8606
p=0.3315
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

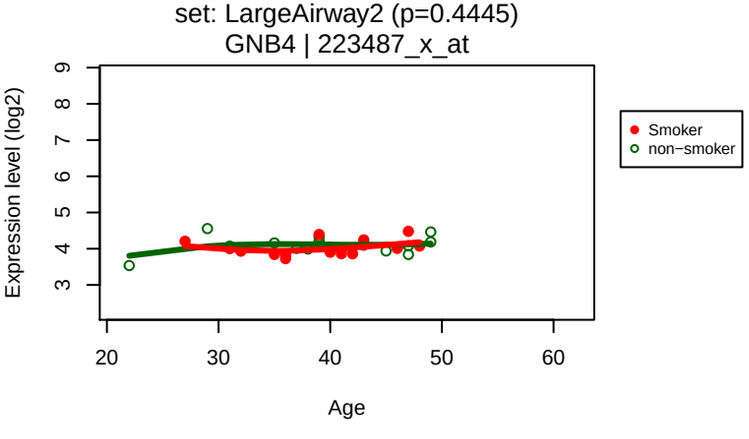

Gene expression ratio (smokers/nonsmokers)
gene: GNB4 | 223487_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7203
p=0.2219
p=0.8221
p=0.2535
p=0.4445
p=0.0726
p=0.8606
p=0.3315
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

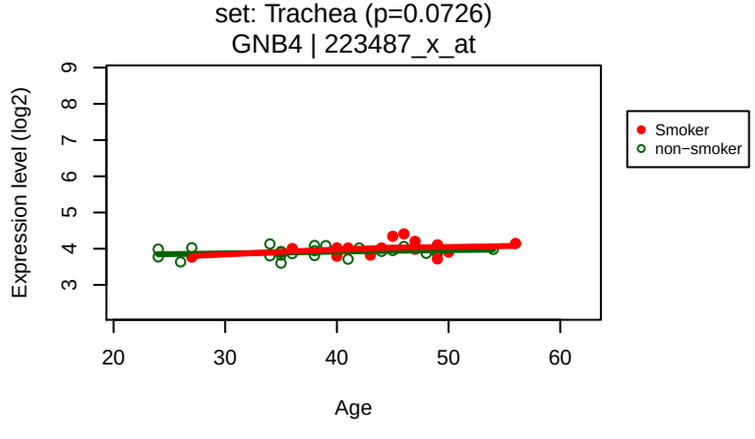

Gene expression ratio in Males
gene: GNB4 | 223487_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5618
p=0.327
p=0.151
p=0.8231
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNB4 | 223487_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3323
p=0.826
p=0.5741
p=0.1784
SmallAirway2
LargeAirway2
Trachea
Macrophage2

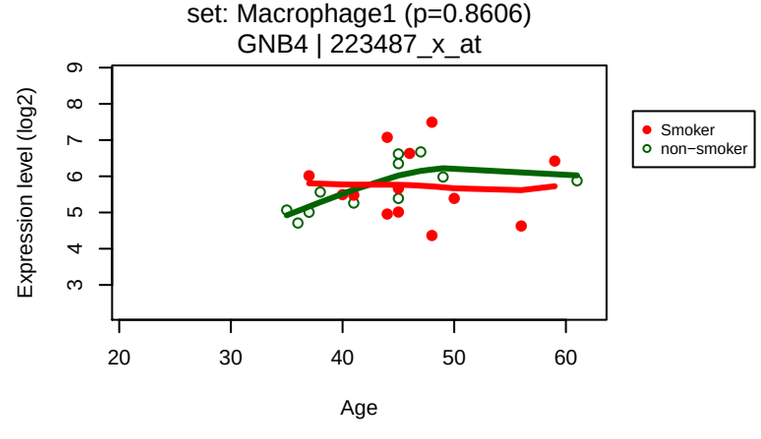

set: SmallAirway1 (p=0.7203)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker

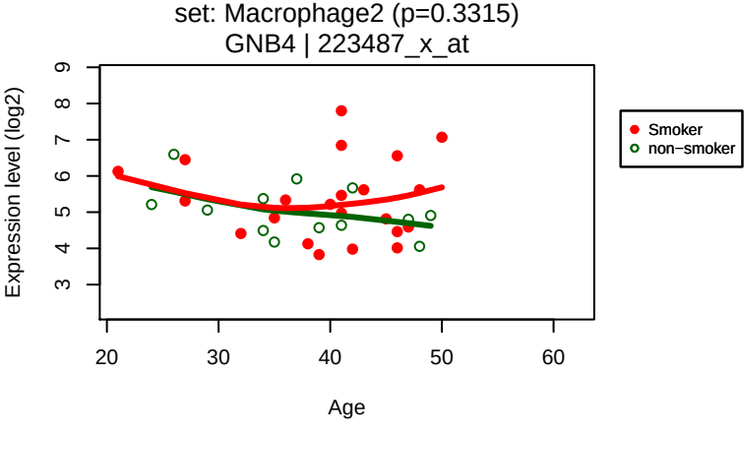

set: SmallAirway2 (p=0.2219)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker

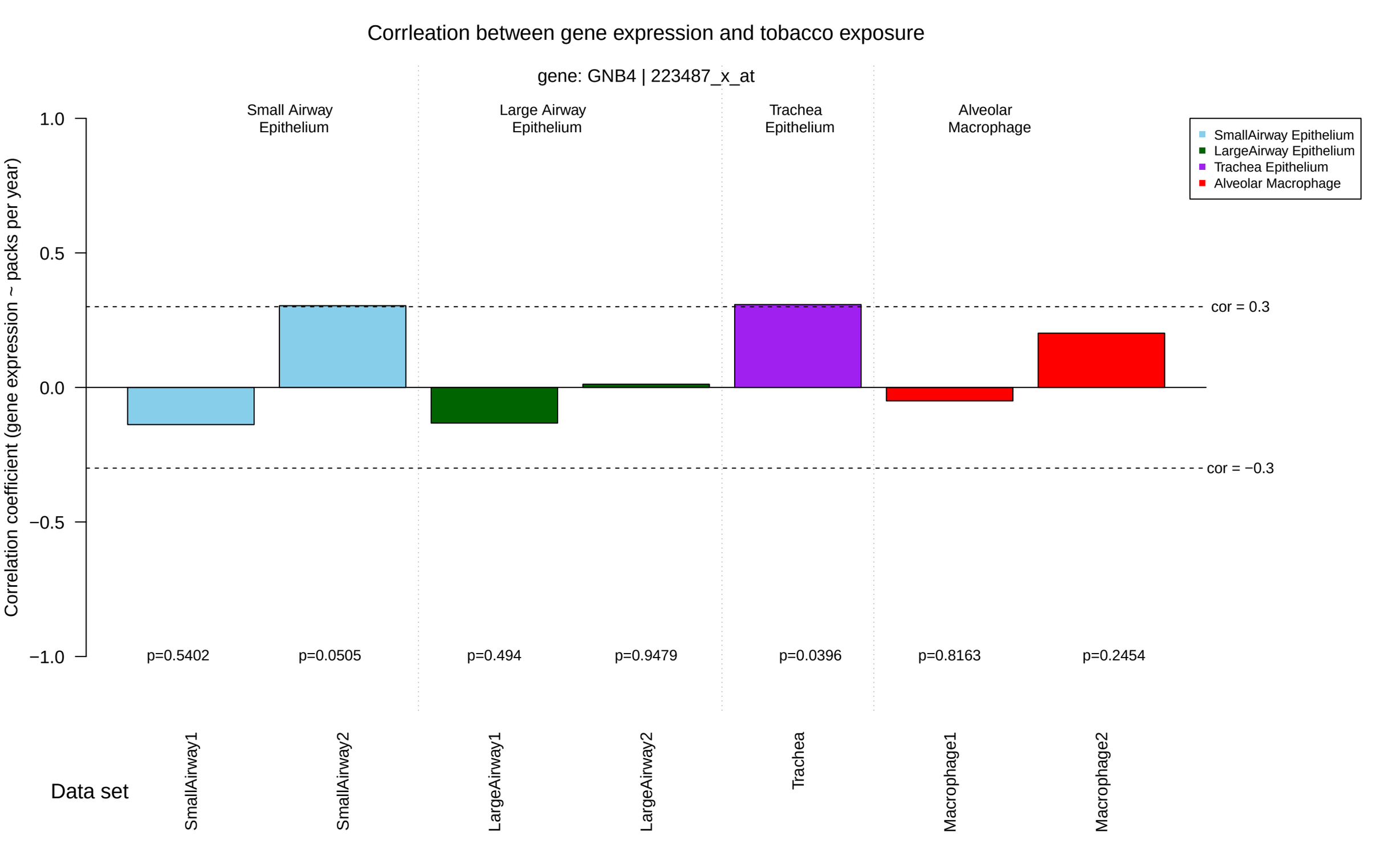

set: LargeAirway1 (p=0.2535)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4445)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0726)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8606)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3315)
GNB4 | 223487_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNB4 | 223487_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5402
p=0.0505
p=0.494
p=0.9479
p=0.0396
p=0.8163
p=0.2454
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2